

Session 13

B1

Output:

0

False

Then an `InvalidOperationException` is thrown with message "Sequence contains no elements".

Reason: `Count()` on an empty list returns 0, `Any()` returns False, while `Average()` cannot calculate a value for an empty sequence.

B2

Expected output:

2 3 1 4

Reason: `GroupBy()` preserves the order in which each key first appears in the source sequence. It does not automatically sort the groups.

B3

Expected output:

3

Reason: `Where()` uses deferred execution. The query is not executed until `Count()` is called, so the newly added student is included in the result.

Part F : database migration

```
Package Manager Console
Package source: All
Default project: StudentPortalConsole
PM> Get-Command Add-Migration

CommandType      Name                Version      Source
-----
Function         Add-Migration       10.0.10      EntityFrameworkCore

PM> Add-Migration InitialCreate
Build started...
Build succeeded.
8/1/2026 11:30:50.278 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infrastructure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this mode should only be
enabled during development.
To undo this action, use Remove-Migration.
PM> Update-database
Build started...
Build succeeded.
8/1/2026 11:31:12.654 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infrastructure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this mode should only be
enabled during development.
8/1/2026 11:31:13.061 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (112ms) [Parameters=[], CommandType='Text', CommandTimeout='60']
```

Data Retrieve :

```
Microsoft Visual Studio Debug Console
ructure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this
mode should only be enabled during development.
info: 8/2/2026 01:31:14.797 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (86ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
FROM [Students] AS [s]
Yara Adel - Year 2 - GPA 3.5
Omar Hesham - Year 3 - GPA 2.8
Nada Samir - Year 1 - GPA 3.9
Kareem Fouad - Year 4 - GPA 3.2
info: 8/2/2026 01:31:14.900 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (11ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
FROM [Students] AS [s]
WHERE [s].[Gpa] > 3.0E0
info: 8/2/2026 01:31:14.903 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (2ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
FROM [Students] AS [s]
info: 8/2/2026 01:31:14.928 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (17ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [s].[FullName]
FROM [Students] AS [s]
WHERE [s].[Gpa] > 3.0E0
ORDER BY [s].[Gpa] DESC
Nada Samir
Yara Adel
Kareem Fouad
```

Migration Analysis:

1.

The Up() method will create the following tables:

- Students
 - Instructors
 - Courses
-

2.

- **Gpa:** float
 - **FullName:** nvarchar(max)
-

3.

No

FullName is **not nullable**.

I did not explicitly configure it. Entity Framework Core inferred that FullName is required because it is defined as a **non-nullable string** in the model.

4.

The Down() method would remove all the tables created by the Up() method (Students, Instructors, and Courses), effectively rolling back the migration.

SQL data base created with right tables :

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the Object Explorer shows the database structure for 'localhost (SQL Server 17.0.1000.7 - YOUSSEFEZZAT\youssefezzat)'. The 'Databases' folder is expanded, showing 'ITI_StudentPortal' selected. Under 'ITI_StudentPortal', the 'Tables' folder is expanded, listing several tables including 'dbo.Students'. The main pane on the right shows a SQL query executed in the 'SQLQuery8.sql...ot connected' window. The query is:

```
1 SELECT TOP (1000) [Id]
2     ,[FullName]
3     ,[YearOfStudy]
4     ,[Gpa]
5 FROM [ITI_StudentPortal].[dbo].[Students]
6
```

Below the query, the 'Results' tab shows the output of the query. The results are displayed in a table with 5 columns: 'Id', 'FullName', 'YearOfStudy', and 'Gpa'. The table contains 4 rows of data:

	Id	FullName	YearOfStudy	Gpa
1	1	Yara Adel	2	3.5
2	2	Omar Hesham	3	2.8
3	3	Nada Samir	1	3.9
4	4	Kareem Fouad	4	3.2